Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering fluorescent structures that overlap in projection requires measurements that distinguish their axial arrangements. We encode depth in the illumination of a fixed-focus fluorescence microscope by scanning a plane wave in angle through a fixed, defocused phase plate. We learn the plate through the optical forward model to reduce correlations between depth-dependent measurement signatures, without specimen examples or a reconstruction decoder. We compare the learned plate with an untrained random-phase plate on simulated measurements of 60 mouse pyramidal neuron crops, at eightfold measurement compression and matched nominal excitation dose. Mean axial-profile Wasserstein-1 (W1) error decreases from 0.37 *to* $0.22\ \mu\mathrm{m}$ *with total variation (TV), from* 0.58 *to* $0.29\ \mu\mathrm{m}$ *with pretrained plug-and-play (PnP), and from* 0.27 *to* $0.20\ \mu\mathrm{m}$ *with neuron-trained PnP. The gain extends to a feed-forward decoder trained separately for each plate. Controlled crossing-filament tests show improved axial separation under two of the three iterative methods. Improvements over the untrained plate persist with background and read noise, and at reduced dose. Our plate also gives lower error than the tested coherence- and Fisher-information-based designs. A plate trained end-to-end with a decoder on specimen examples differs from ours by less than* $0.01\ \mu\mathrm{m}$ *in mean error under the iterative methods; its more extensively trained decoder achieves* $0.03\ \mu\mathrm{m}$ *lower error. These results show that learning depth-discriminative illumination independently of a decoder improves axial recovery across the tested reconstruction methods.*

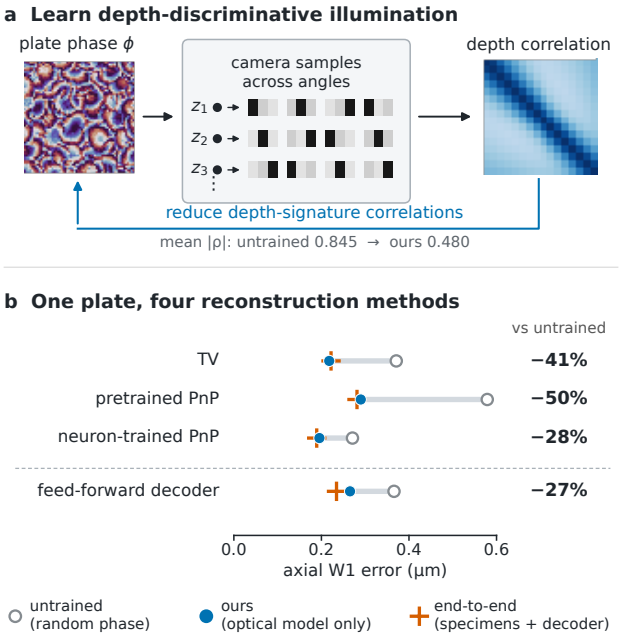


Figure 1. **Decoder-free optical learning improves axial recovery across four reconstruction methods.** (a) Training decorrelates detector-integrated depth signatures. Strips are schematic; the matrix shows our plate. Mean absolute off-diagonal correlations use 32 directions and 44×44 interior positions. (b) Mean W1 on simulated measurements at eightfold measurement compression ($C = 8$) and matched nominal dose. Percentages compare our plate with the untrained plate. Legend entries identify the inputs used for optical training. Feed-forward networks are fine-tuned for each plate, including the end-to-end plate. Paired intervals are in the supplement.

1. Introduction

Fluorescent branches that overlap in a camera image may lie at different depths. In fixed-focus fluorescence microscopy, a broad detection depth of field can make different axial arrangements produce similar images. Reconstruction then relies more heavily on prior assumptions [3, 11]. An accurate mean depth can still conceal two structures reconstructed as one. Faithful recovery therefore requires measurements that distinguish how fluorescence is distributed along depth.

We encode depth in the excitation by scanning the incident angle of a plane wave through a fixed, defocused phase plate (Fig. 2). Propagation produces structured illumination patterns that shift by different amounts at different depths as the angle changes. Structures at different axial positions therefore experience different sequences of excitation intensities. A camera records one fluorescence image per angle while the plate, specimen and detection focus remain stationary. A random phase plate already supports volumetric

reconstruction, but the phase profile determines how clearly the measurements distinguish axial structures.

We learn the phase profile to make the measurement responses of different depths more distinguishable. A voxel's measurement signature collects its camera responses across illumination angles, accounting for excitation, detection blur and pixel integration. We minimize weighted squared correlations between these signatures at different depths but the same lateral position (Fig. 1a). Training uses only the optical forward model, without specimen examples or a reconstruction network. The plate can therefore be designed before a reconstruction method is chosen.

Fisher information and worst-case coherence are established criteria for designing optics independently of a decoder [8, 26]. They emphasize sensitivity to small depth changes and the most similar pair of measurement responses, respectively. We compare our objective with these criteria under matched optimization budgets. We also evaluate end-to-end optical learning, which optimizes the phase plate jointly with a reconstruction network using specimen examples [5, 20].

We evaluate the optical designs on simulated measurements of mouse pyramidal neuron reference volumes from the public DEEP^2 dataset [29, 30]. Reconstruction uses total variation (TV), plug-and-play (PnP) with pretrained or neuron-trained denoisers, and a feed-forward decoder trained separately for each plate (Fig. 1b). Alongside intensity fidelity, we assess mean-depth placement and the full axial distribution of fluorescence. Neuronal examples illustrate the recovered profiles, while controlled filament crossings test the separation of overlapping structures.

Our contributions are threefold:

1. **Depth-specific optical learning.** We learn a defocused phase plate by minimizing weighted squared correlations between detector-integrated signatures of different depths at the same lateral position, without specimen examples or a reconstruction network.
2. **Competitive recovery across reconstruction methods.** At eightfold measurement compression and matched nominal excitation dose, our plate achieves 10–26 % lower mean axial-profile Wasserstein-1 error than Fisher-information and worst-case-coherence designs across four reconstruction methods. Under iterative reconstruction, mean error is close to that of the end-to-end plate.
3. **Improved axial structure recovery.** Relative to the untrained plate, axial-profile error decreases by 27–50 % across these methods. Controlled crossing tests also demonstrate improved filament separation under two of the three iterative methods.

2. Related work

Depth encoding in fluorescence microscopy. Depth can be encoded through detection optics or structured excitation. Light-field microscopy, DiffuserCam and Miniscope3D exploit depth-dependent detection responses [1, 4, 31]. On the excitation side, Pascucci *et al*. [21] recover volumes from lateral speckle scans through axial decorrelation, with saturation improving resolution, while Yeh *et al*. [32] combine translated speckles with axial scanning. Yilmaz *et al*. [33] use incident-angle scanning through a fixed scattering layer for two-dimensional fluorescence recovery; related work obtains depth cues from angular correlations [27] and tilted-pattern shifts [13]. These precedents motivate our random phase-plate baseline. We optimize the phase profile of a defocused plate for depth discrimination, keeping the plate fixed throughout acquisition.

Learning optical encoders. End-to-end approaches jointly optimize optics and reconstruction networks [12], including DeepSTORM3D [20] for emitter localization and FourierNets [6] for volumetric microscopy. Kellman *et al*. [16] optimize illumination for a specified reconstruction algorithm. Decoder-free design instead targets measurement properties. Shechtman *et al*. [26] use Fisher information to design point-spread functions (PSFs) for three-dimensional localization. Miniscope3D [31] aggregates cross-coherence between depth-dependent detection PSFs, using a smoothed maximum over lateral shifts for each depth pair. Squared-correlation objectives have a classical frame-potential interpretation [2], and related Gram-matrix criteria are established in sensing-matrix design [7]. IDEAL [23] and IDEAL-IO [19] optimize estimated mutual information using fitted measurement distributions, without a reconstruction decoder. Our objective specializes coherence-based design to detector-integrated voxel signatures across excitation angles, comparing depths at the same lateral position. Training uses the optical model alone, without specimen examples or a fitted measurement distribution.

Volumetric reconstruction and priors. Reconstruction methods range from total variation regularization [25] to plug-and-play denoisers [28] and learned networks following a model-based inverse [14]. Multi-slice fusion [17] combines lower-dimensional denoisers along complementary orientations of a shared volume. We use these complementary approaches to assess whether gains from learned illumination persist across different reconstruction priors.

3. Method

3.1. Depth-encoding acquisition

A plane wave illuminates a fixed phase plate with transmission $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagation converts its phase modulation into spatially structured excitation (Fig. 2a). Changing the illumination angle shifts the excitation pattern by an

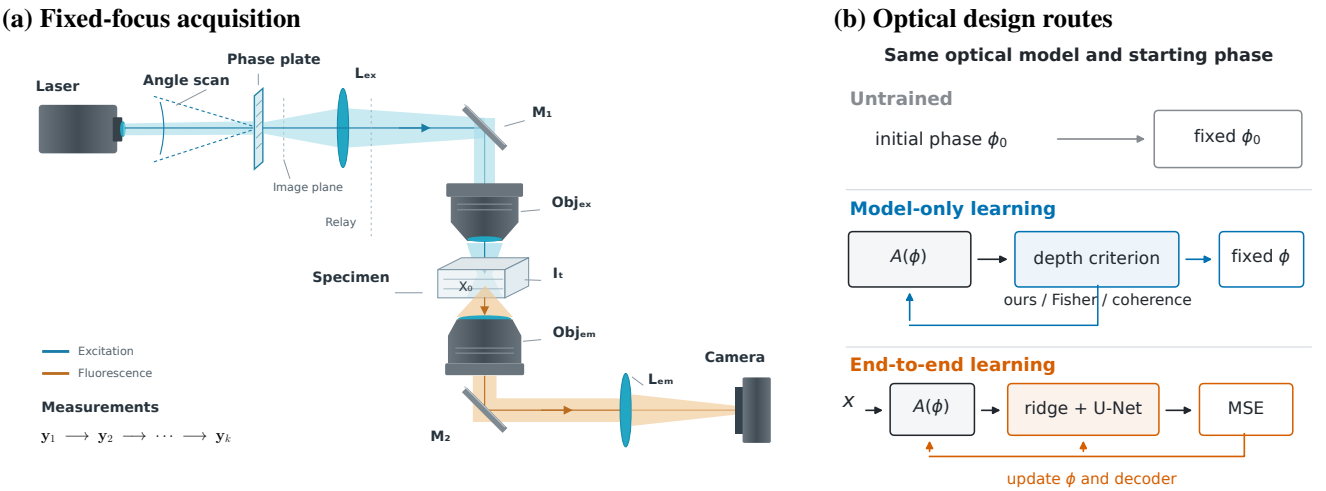


Figure 2. **A shared optical model supports three design routes.** (a) Only the incident angle changes during acquisition; the plate, specimen and detection focus remain stationary. Blue and orange indicate excitation and emission. (b) The untrained phase is retained, optimized using a model-only criterion, or optimized jointly with a decoder on specimen volumes. Each route yields a distinct fixed plate for reconstruction. Shared optics do not imply equal training budgets.

amount that varies with depth. In the ideal paraxial thin-screen approximation, a change $\Delta\mathbf{s}$ in transverse direction cosine produces a lateral displacement $\Delta\mathbf{r}(z) \approx d(z)\Delta\mathbf{s}$, where $d(z)$ is the axial distance from the plane conjugate to the plate [9, 10]. Plate defocus places this shear pivot outside the specimen, producing depth-dependent excitation sequences as the angle is scanned. We compute the actual excitation using a vectorial relay model that accounts for the high numerical aperture (NA), finite pupil and departures from pure translation [18, 24]. Acquisition is sequential: one fluorescence image is recorded per angle, while the plate, specimen and detection focus remain fixed.

Image formation and signatures. Let $X_0(\mathbf{r}, z)$ denote fluorescence response per unit excitation, with $\mathbf{r} = (x, y)$. Under linear, unsaturated fluorescence, illumination $I_t(\mathbf{r}, z; \phi)$ produces an emission source proportional to $X_0 I_t$. Contributions from different locations add incoherently through the detection optics. For a piecewise-constant reconstruction with coefficient x_j in voxel V_j, the noiseless model is

$$\bar{\mathbf{y}} = \mathsf{A}_\phi \mathbf{x},$$
$$(\mathsf{A}_\phi)_{(t,p),j} = \int_{V_j} I_t(\mathbf{r}, z; \phi)\, g_p(\mathbf{r}, z)\, \mathrm{d}^2\mathbf{r}\, \mathrm{d}z, \tag{1}$$

where $g_p(\mathbf{r}, z)$ is the response of camera pixel p to unit emission at $(\mathbf{r}, z)$, including the laterally shift-invariant emission point-spread function and pixel averaging [32]. Excitation modulates the specimen before detection blur, so the camera need not resolve the illumination pattern itself. We evaluate the voxel integrals on a finer optical grid to retain within-voxel variation; fine-grid measurement generation and specimen discretization error are detailed in the supplement. Each column of A_ϕ collects one voxel's responses across illumination angles and camera pixels, defining its measurement signature.

3.2. Phase-plate learning

Depths at the same lateral position have overlapping detection responses, so their excitation sequences provide an important source of axial discrimination. We optimize the phase plate to reduce correlations between the corresponding measurement signatures. For sampled illumination angles $\mathcal{T}$, let $\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z}$ denote the signature at lateral position $\mathbf{r}$ and depth z, centered over all angle–pixel entries and normalized to unit norm. The inner product $\rho_{zz'} = (\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z})^\top \mathbf{a}^{\mathcal{T}}_{\mathbf{r},z'}$ measures correlation between two depths at that position.

We minimize the weighted mean squared correlation, averaged over lateral positions and sampled angle sets:

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}}\left[\frac{\sum_{z\neq z'} w_{zz'}\, \rho^2_{zz'}}{\sum_{z\neq z'} w_{zz'}}\right]. \tag{2}$$

The depth-pair weights $w_{zz'}$ are specified in the supplement. Squaring penalizes correlations of either sign. With uniform weights, the numerator is the off-diagonal energy of the depth-signature Gram matrix, equivalent to the classical frame potential up to a constant [2]. The supplement gives a statistical interpretation of the unweighted criterion.

Training uses the optical forward model alone, without specimen examples or a reconstruction solver. We optimize the phase map by backpropagating through excitation propagation, the vectorial relay and detector integration. Signature normalization prevents uniform power scaling from directly

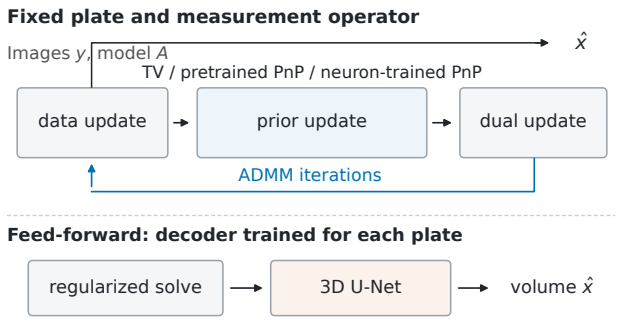


Figure 3. **Reconstruction is evaluated after the plate is fixed.** TV and PnP use the alternating direction method of multipliers (ADMM) with different prior updates. The feed-forward route uses a regularized solve followed by a 3D U-Net whose weights are trained separately for each plate. Full updates are in the supplement.

lowering the loss; nominal excitation dose is controlled separately during evaluation.

Alternative decoder-free objectives. We compare our objective with two established design criteria: worst-case coherence [8] and depth Fisher information [26]. The coherence objective targets the most strongly correlated depth pair at each lateral position through a smooth maximum of $|\rho_{zz'}|$. The Fisher objective minimizes $-\mathbb{E}_z \log \mathcal{I}_z$, where $\mathcal{I}_z$ measures local sensitivity to axial displacement under a Poisson model with unknown emitter brightness. Both are optimized using the same optical model and phase parameterization. Their full formulations and shared optimization settings are provided in the supplement.

End-to-end counterpart. Using the same optical model, we jointly optimize the plate and a decoder on neuronal training volumes (Fig. 2b). The decoder applies a regularized linear solve followed by a residual 3D U-Net. Training minimizes reconstruction mean squared error and updates both the phase map and decoder parameters. We then fix the learned plate and evaluate it with all reconstruction methods. The full decoder formulation and training protocol are provided in the supplement.

3.3. Reconstruction with fixed optics

We evaluate three iterative methods and a feed-forward decoder using the operator A_ϕ of each fixed plate (Fig. 3). TV combines a quadratic measurement term with isotropic 3D total variation and $[0, 1]$ intensity bounds [25]. PnP uses denoisers as implicit priors within ADMM [15, 28]. Following multi-slice fusion [17], pretrained PnP applies a natural-image DnCNN independently along three orthogonal slice orientations of a shared volume [34]. For slice denoising, neuron-trained PnP uses a 2D network trained on neuronal volumes.

The TV regularizer's form and PnP denoiser weights remain fixed across plates. For these iterative methods, scalar parameters are selected separately for each plate and acquisition condition using held-out validation volumes. Feed-forward decoders share an architecture, with weights fine-tuned separately for each fixed plate. Full solver updates, denoiser configurations and training protocols are provided in the supplement.

4. Experimental setup

Simulation settings. We simulate linear fluorescence in a homogeneous, non-scattering medium. Excitation and emission use a common wavelength of $532\,\mathrm{nm}$ as a modeling approximation, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has $0.32\,\mu\mathrm{m}$ features and lies $6.6\,\mu\mathrm{m}$ before the intermediate illumination image plane in specimen-equivalent coordinates. Fields use $0.08\,\mu\mathrm{m}$ lateral and $0.32\,\mu\mathrm{m}$ axial sampling, binned to a $0.32\,\mu\mathrm{m}$ analysis grid. Measurements are generated on this grid before averaging the reference into $0.64\,\mu\mathrm{m}$ reconstruction voxels, retaining within-voxel specimen variation.

Volumes and compression. A reconstruction has $16 \times 48 \times 48$ voxels ($N = 36{,}864$), covering $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image contains 12×12 samples. Acquiring $k = 256$, 64 or 32 directions gives measurement compression $C = N/(144k) = 1$, 4 or 8. For $C > 1$, five randomly drawn direction subsets are paired across encoders.

Data and validation. References are public two-photon stacks of mouse pyramidal neurons used in DEEP^2 [30]. We normalize each crop, resample it laterally and assign its 32 depth planes the $0.32\,\mu\mathrm{m}$ analysis spacing, compressing the original axial extent about threefold. All reported distances use simulation coordinates. Six cells from three animals contribute 60 test crops: two per cell selected by reference structure before reconstruction, and eight sampled at random after reconstruction parameters were fixed. Other full-field studies use the 12 prespecified crops unless stated otherwise. Neuronal denoisers and supervised optical/decoder training use 459 crops from a fourth animal. The supervised models use 48 validation crops from a held-out stack of that animal. TV/PnP scalar parameters are selected separately for each encoder and acquisition condition on two vascular crops from another animal. Test volumes are excluded from training, selection and tuning.

Optical design controls. The untrained plate has independent phases drawn uniformly on $[0, 2\pi)$ at each feature and

is the common starting phase for all learned plates. Our criterion, depth Fisher information and smoothed worst-case coherence share the phase parameterization, initialization, sampled positions/angles and 1000-step Adam budget. The end-to-end counterpart uses the same optics and initial phase, with a separate supervised training schedule. Feed-forward decoders share the warm-up and final fine-tuning recipe; the end-to-end decoder also receives joint-training updates (Sec. 5.5). Thus, acquisition and physical models are matched, while total decoder training differs. Additional controls include random phase realizations, defocused binary masks, a plate-free scan and an in-focus digital micromirror device (DMD); full protocols are in the supplement.

Photon budget. All encoders are normalized to the same mean excitation over the specimen and all 256 directions. With k acquired images, camera counts follow $n_i \sim \mathrm{Poisson}(\kappa_0\, 256 \bar{y}_i/k)$, with normalized data $y_i = n_i/(\kappa_0\, 256/k)$ for reconstruction. For each specimen, κ_0 is set using the untrained plate to give 1000 expected photons per camera pixel per image at the mean signal for $k = 256$, then shared across encoders. This matches deposited dose for the full scan and in expectation over direction subsets; we call it *matched nominal dose*. Detected counts can differ. Mean-dose matching does not constrain excitation peaks; the model assumes an unsaturated response throughout. Fewer images receive longer exposure per image. Apertures, background/read noise and reduced-dose conditions are specified in the supplement.

Metrics and statistics. Our primary metric compares full axial intensity profiles using one-dimensional Wasserstein-1 distance [22]: the mean distance that unit-mass signal must move along depth to match the reference profile. For each reference-signal column c, normalize its reference and nonnegative reconstructed profiles, with CDFs F_c and $\hat{F}_c$. Then

$$E_{\mathrm{W1}} = \sum_c \pi_c\, \Delta z \sum_{i=0}^{Z-2} |F_c(i) - \hat{F}_c(i)|, \qquad (3)$$

where $Z = 16$, $\Delta z = 0.64\,\mu\mathrm{m}$ and π_c is the normalized reference column brightness. Columns must exceed 10 % of the maximum reference column brightness. Centroid error uses the same columns and weights. PSNR and foreground SSIM complement these normalized axial metrics with intensity and image-quality assessment. We average draws within each crop and report paired changes in crop means. Confidence intervals resample the six cells with their crops intact (10,000 replicates), accounting for within-cell dependence. Phantom intervals resample phantom cases after averaging draws.

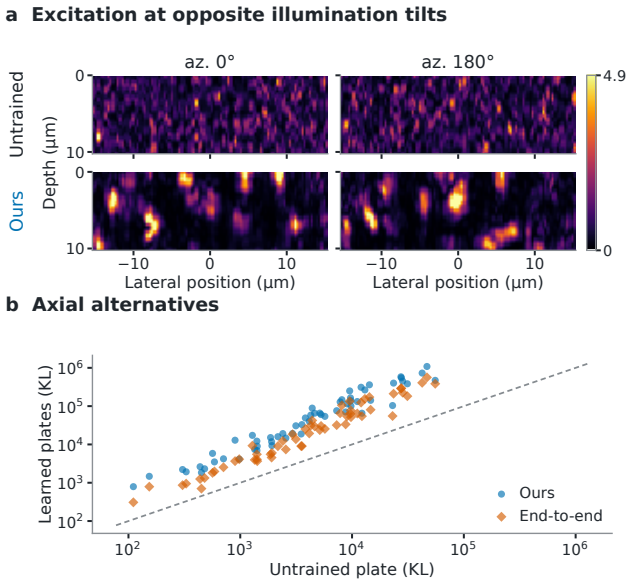


Figure 4. **Optical learning changes the depth code before reconstruction.** (a) Illustrative excitation sections at equal and opposite tilts (azimuths 0° and 180°) in the same central vertical plane, for the untrained plate (top) and ours (bottom). The shared linear intensity scale is relative to the untrained mean. (b) Poisson measurement divergence from merged-depth alternatives for 59 eligible neuronal crops. Points average the five direction subsets; the diagonal indicates equal divergence.

Axial structure tests. Crossing filaments use three lateral geometries, two axial centers and five measurement draws. Success requires matched axial peaks in columns carrying at least half the reference crossing signal, together with correct ordering of the arms. The smallest resolved gap must achieve 80 % success at each center and all larger tested gaps. Separately, measurement discrimination is assessed against merged-depth alternatives preserving the reference projection and column centroids. Definitions and selection rules are in the supplement.

5. Results

5.1. Depth discrimination before reconstruction

Learning makes the defocused plate's depth code more selective (Fig. 4a): strong responses occupy different axial regions as the incident angle changes. Over the interior field, mean absolute depth-signature correlation falls from 0.845 to 0.480 (Fig. 1a); the end-to-end plate reaches 0.628 (matrices in the supplement). These correlations use the full detector-integrated signatures, rather than illumination intensity alone.

We test whether measurements distinguish each neuronal reference from a version whose depth layers are merged while its projection and column centroids remain unchanged.

Poisson Kullback–Leibler (KL) divergence measures separation between the two expected-count distributions, after fitting the alternative's global brightness. Our plate increases this divergence by a geometric mean factor of 10.0 relative to the untrained plate (95 % CI: 8.7–11.8), and 2.0 relative to the end-to-end plate (1.9–2.1). Both increases hold for all 59 eligible crops after averaging direction subsets (Fig. 4b). This tests discrimination of specific axial alternatives, independently of a reconstruction prior; it does not establish unique recovery of arbitrary volumes.

5.2. One learned plate, four reconstruction methods

Our plate lowers mean axial-profile W1 by 41 % with TV, 50 % with pretrained PnP, 28 % with neuron-trained PnP and 27 % with the feed-forward decoder, relative to the untrained plate (Fig. 1b, Tab. 1). Paired cell-bootstrap intervals exclude zero for all four comparisons. W1 improves on all 60 crops under each TV/PnP method. The improvements persist on the prespecified and later randomly selected crops, and a second optical sampling seed gives similar results from the same phase initialization.

Centroid errors also decrease, by 42 %, 54 % and 30 % under TV and the two PnP methods, and PSNR increases by 1.4–1.8 dB across the four methods. Thus, profile recovery improves alongside mean depth placement and image quality. With a non-learned TV prior, our plate also gives lower W1 than the untrained plate with neuron-trained PnP (0.218 versus $0.271\,\mu\mathrm{m}$).

Figure 5 shows a neuronal crop where both learned plates reduce crop-wide profile error. At the marked column, two unequal axial peaks share a lateral position: our plate retains the weaker component more faithfully, while the untrained and end-to-end reconstructions attenuate it. This local W1 advantage persists across all five direction subsets, although the two learned plates have similar crop-wide W1. Reference-selected views from two animals and all three iterative methods are provided in the supplement.

For context, we test the same priors with random binary patterns projected in focus by a DMD. On the 12 prespecified crops, this control gives W1 of 2.24–$2.38\,\mu\mathrm{m}$ across TV/PnP, near the $2.27\,\mu\mathrm{m}$ of an estimate that spreads each column uniformly over depth. This behavior concerns the tested in-focus encoder: defocused amplitude masks do recover depth and remain competitive with untrained phase plates in centroid error. In a separate 12-crop control with pretrained PnP, our starting plate lies within the centroid-error range of three untrained phase realizations.

5.3. Recovery of overlapping structures

Crossing filaments directly test two structures that overlap in projection but lie at different depths. In this controlled task, the smallest gap meeting the resolution-and-ordering criterion falls from 3.84 to $2.56\,\mu\mathrm{m}$ with our plate under TV and pretrained PnP (Fig. 6). It remains $3.84\,\mu\mathrm{m}$ under neuron-trained PnP despite improved neuronal W1. The zero-gap control shows no false splitting.

The supplement reports full crossing curves and extensions with weaker and additional filaments.

5.4. Comparison of model-only objectives

All three plates learned from model-only criteria improve upon the untrained plate (Tab. 1). The plate trained with our criterion gives 12–26 % lower W1 than the worst-case-coherence plate and 10–21 % lower than the depth-Fisher plate across the four reconstruction methods. Paired cell-bootstrap intervals exclude zero in each primary comparison. These results compare the stated implementations under matched sampling and optimization budgets, with four angles sampled per update. Recorded training traces are provided in the supplement. The TV/PnP gains persist with a second optical sampling seed on the same 60 crops, including a seed-matched comparison with our second plate; full results are in the supplement. On the 12 prespecified crops, an uncentered squared-correlation variant gives nearly identical recovery under neuron-trained PnP (0.207 versus $0.206\,\mu\mathrm{m}$), so centering is not an established source of the advantage.

5.5. Comparison with joint optical–decoder training

The two learned plates differ by less than $0.01\,\mu\mathrm{m}$ in mean W1 under the three iterative methods (Tab. 1). The end-to-end plate has 1.9 % higher mean W1 under TV, 3.0 % lower under pretrained PnP and 3.2 % lower under neuron-trained PnP. Paired intervals exclude zero in the stated directions; absolute intervals are reported in the supplement. Centroid error also favors end-to-end by 5.4 % under both PnP methods, with no established TV difference. Both plates improve substantially over the untrained plate; the end-to-end design also transfers beyond its training decoder. It reaches this accuracy with less-decorrelated signatures (mean absolute correlation 0.628 versus 0.480), showing that the diagnostic alone does not rank reconstruction quality.

With separately fine-tuned feed-forward decoders, the end-to-end system achieves lower W1 (0.234 versus $0.265\,\mu\mathrm{m}$, a 12 % reduction) and 0.48 dB higher PSNR. After a shared 5000-update warm-up, its decoder receives 7500 joint-training updates and 3000 fixed-plate updates, compared with 3000 fixed-plate updates for ours. The untrained plate's decoder receives all 8000 updates on its own operator. This compares trained optical–decoder systems with unequal decoder histories. The TV/PnP comparison holds regularizer families and denoiser weights fixed and shows that optical-model-only training attains close recovery without specimen examples or reconstruction feedback.

Additional acquisition conditions give a mixed compar-

		Model-only optical training			Joint training
Reconstruction	Untrained	Depth Fisher	Worst-case coherence	Ours	End-to-end
TV	0.371	0.251	0.264	**0.218**	0.222
Pretrained PnP	0.579	0.367	0.390	0.290	**0.281**
Neuron-trained PnP	0.271	0.218	0.223	0.195	**0.189**
Feed-forward	0.366	0.314	0.314	0.265	**0.234**

Table 1. **Optical design evaluated across four reconstruction methods.** Mean axial-profile W1 (μm) on simulated measurements at $C = 8$ and matched nominal dose: 60 crops from six cells and three animals, averaging five direction subsets per crop. Model-only criteria use sampling seed 0 and matched optimization settings; bold and underline mark the lowest and second-lowest means, respectively. The feed-forward network is fine-tuned separately for each plate. All trained plates start from the untrained phase map. Paired intervals and intensity metrics are in the supplement.

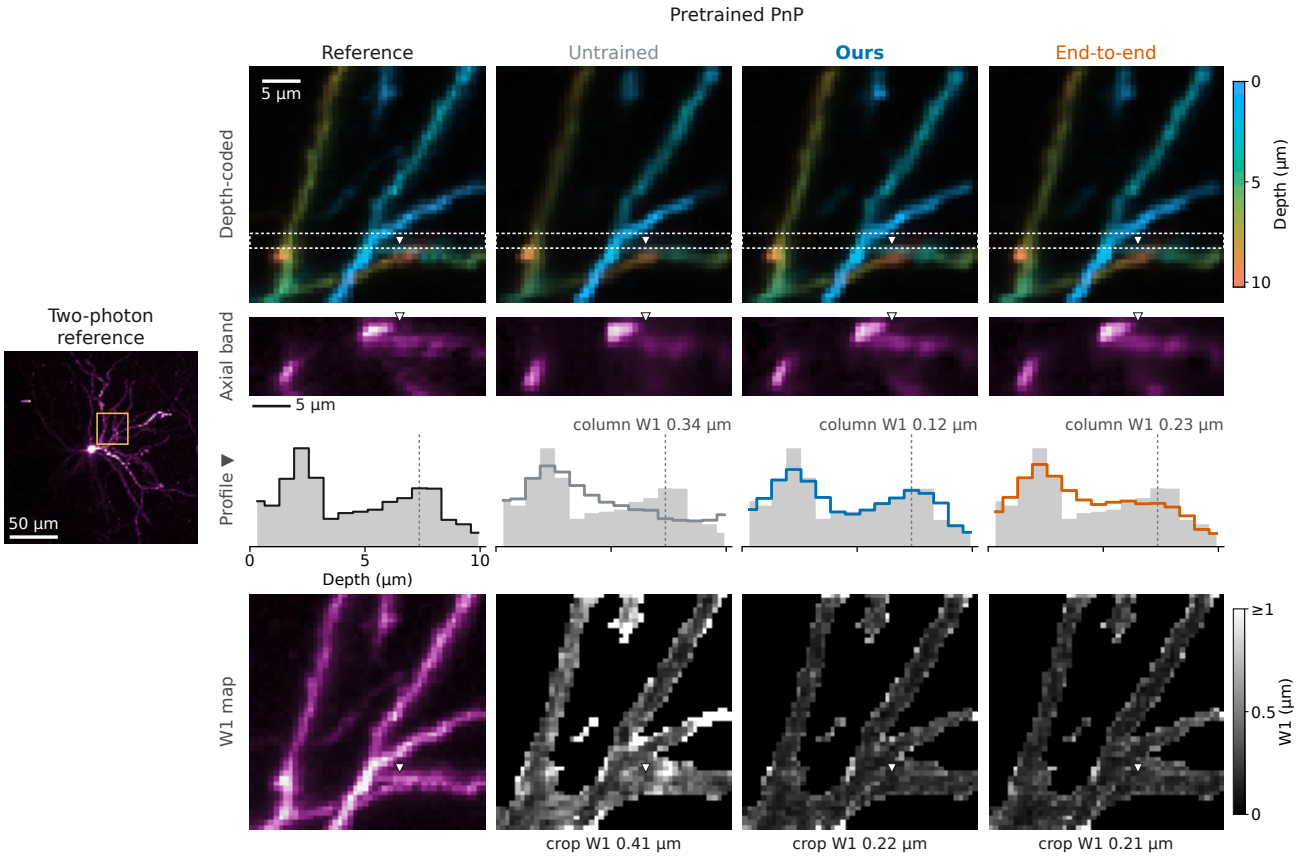


Figure 5. **Recovery of unequal axial peaks.** Matched reconstructions with pretrained PnP at $C = 8$. The marked column, chosen after inspecting reconstructions, illustrates better retention of the weaker peak by our plate (dashed depth line). Profiles have unit mass, with the reference in gray; column and crop W1 are labeled separately. The bottom-left panel shows reference intensity. Fluorescence images use magenta; W1 maps run from black (0) to white ($\geq 1\ \mu$m), with black also marking pixels outside the reference signal mask. Reconstruction images share a linear intensity scale and noise draw. Reconstructed views use simulation coordinates; the two-photon context uses tissue coordinates. Five-subset profiles and selection details are in the supplement.

ison. With 256 images, our plate has lower W1 under pretrained PnP; differences under TV and neuron-trained PnP are not established. At one tenth of the dose, the end-to-end plate has lower neuronal W1 under neuron-trained PnP (0.227 versus $0.239\ \mu$m), while our plate has 17 and 15 percentage points higher pooled crossing success under TV and pretrained PnP, respectively (Fig. 6c). Differences in the remaining reduced-dose comparisons and all full-dose

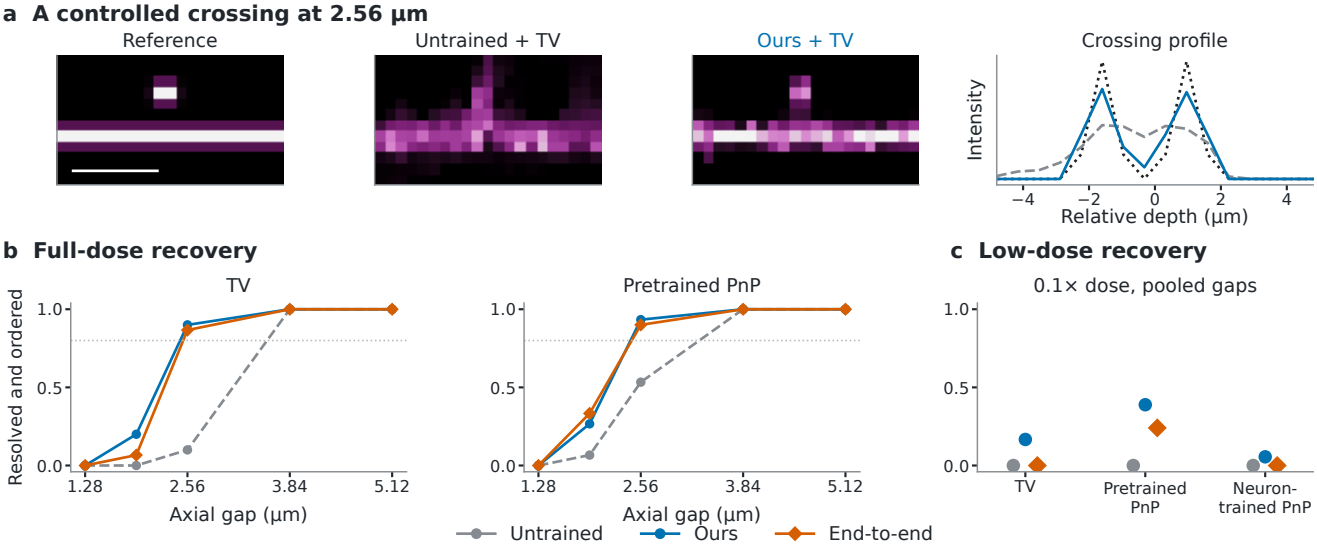


Figure 6. **Controlled structure recovery depends on the code, prior and dose.** (a) Matched fluorescence sections (magenta) and profiles through one crossing at our plate's TV threshold gap; the first fixed geometry, center and draw are shown, with a shared intensity scale. Bar: 5 μm. (b) Full-dose success rates pool the two axial centers; the 80 % criterion is applied to each center separately. (c) Reduced-dose success pooled over all three positive gaps (1.28–2.56 μm), for each iterative method. Full gap curves and paired phantom-bootstrap intervals are in the supplement.

crossing comparisons are not established.

5.6. Acquisition budgets and calibration

On the 12 prespecified crops, learning can reduce the number of images needed for a given recovery level. With 32 images, our plate gives lower W1 than the untrained plate with 256 images under TV (0.196 versus 0.249 μm) and pretrained PnP (0.277 versus 0.413 μm). Neuron-trained PnP gives close means (0.206 versus 0.203 μm). These comparisons use the same nominal dose, with longer exposure per image for the shorter scan. At 0.3 times dose, our plate reduces W1 by 22–39 % relative to the untrained plate at full dose. Gains also persist with background and read noise. Full budget curves and paired comparisons are in the supplement.

Calibration is important for preserving these gains. Known plate-distance offsets of ± 0.5 μm retain 27–28 % W1 reductions under neuron-trained PnP after updating the operator and revalidating scalar parameters, without retraining the plate. Unmodeled distance or lateral offsets can reverse the advantage: a one-feature lateral shift makes our plate 11 % worse than the untrained control (95 % CI: 2–23). Learned encoding therefore requires registration of the realized optics.

6. Discussion and limitations

Optical design before decoder choice. A plate can be designed from the optical model before the reconstruction method is chosen. In this system, the same learned plate lowers axial-profile error under four methods, with mean W1 differing by less than 0.01 μm from the end-to-end plate under the iterative methods. The larger end-to-end gain accompanies its own, more extensively trained decoder. The optical improvement can also reduce measurement requirements: on the 12 prespecified crops, our 32-image scan outperforms the untrained 256-image scan under TV and pretrained PnP at the same nominal dose.

Measurement discrimination and structure. Our plate has lower depth correlations and greater separation from prescribed merged-depth alternatives than the end-to-end plate, yet these diagnostics do not determine their reconstruction ranking. They target particular axial ambiguities. Crossing tests connect the optical improvement to a controlled structural task, with smaller qualifying gaps under TV and pretrained PnP. The unchanged threshold under neuron-trained PnP and mixed reduced-dose results show that this benefit depends on both structure and reconstruction prior. These tests support better axial placement and separation; they do not certify neuronal connectivity.

Physical implementation. All results use simulations of linear fluorescence in a homogeneous medium, with an ideal thin phase plate and a common excitation/emission wavelength. Our patterns concentrate excitation more strongly, so operating power must keep the fluorescence response unsaturated. Neuronal references are axially rescaled to the

simulation grid. Fine-grid measurement generation retains sub-voxel specimen variation but shares optical physics with reconstruction. A measured system must characterize its realized excitation and detection responses. Updating the operator preserves gains for known alignment errors, while unmodeled registration can reverse them.

Scope of the comparison. The test set contains six cells from three animals, with dependent crops and approximate cell-bootstrap intervals. Two optical sampling seeds share one phase initialization. We test specific coherence and Fisher surrogates and one jointly trained architecture, with unequal decoder training histories and an MSE-trained supervised decoder. Supervised training uses coarse-basis measurements and Gaussian noise, while evaluation retains fine-grid specimens and Poisson counts. The evidence supports comparisons within this protocol; it does not establish an optimum over objectives, starting phases or reconstruction methods.

7. Conclusion

We learn depth-encoding illumination from the optical model alone, without specimen examples or a reconstruction decoder. In fixed-focus fluorescence simulations, the same plate lowers mean axial-profile W1 from 0.27–$0.58\,\mu\mathrm{m}$ to 0.20–$0.29\,\mu\mathrm{m}$ across four reconstruction methods at eight-fold measurement compression. It also separates crossing filaments at smaller gaps under two of the three iterative methods. Mean W1 differs from the end-to-end plate by less than $0.01\,\mu\mathrm{m}$ under the iterative methods; the end-to-end system gives lower error with its own decoder. Depth signatures can therefore guide optical learning while leaving reconstruction choice outside the training loop.

LLM Usage Statement

OpenAI ChatGPT/Codex and Anthropic Claude assisted with experimental planning and code, manuscript and supplement organization, drafting and editing, literature checks, numerical audits of simulation outputs, and figure/table preparation. The authors retain responsibility for the methods, results, citations and final manuscript.

References

[1] Nick Antipa, Grace Kuo, Reinhard Heckel, Ben Mildenhall, Emrah Bostan, Ren Ng, and Laura Waller. DiffuserCam: lensless single-exposure 3D imaging. *Optica*, 5(1):1–9, 2018. 2

[2] John J. Benedetto and Matthew Fickus. Finite normalized tight frames. *Advances in Computational Mathematics*, 18: 357–385, 2003. 2, 3

[3] Sayantan Bhadra, Varun A. Kelkar, Frank J. Brooks, and Mark A. Anastasio. On hallucinations in tomographic image reconstruction. *IEEE Transactions on Medical Imaging*, 40 (11):3249–3260, 2021. 1

[4] Michael Broxton, Logan Grosenick, Samuel Yang, Noy Cohen, Aaron Andalman, Karl Deisseroth, and Marc Levoy. Wave optics theory and 3-D deconvolution for the light field microscope. *Optics Express*, 21(21):25418–25439, 2013. 2

[5] Julie Chang and Gordon Wetzstein. Deep optics for monocular depth estimation and 3D object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10193–10202, 2019. 2

[6] Diptodip Deb, Zhenfei Jiao, Ruth R. Sims, Alex B. Chen, Michael Broxton, Misha B. Ahrens, Kaspar Podgorski, and Srinivas C. Turaga. FourierNets enable the design of highly non-local optical encoders for computational imaging. In *Advances in Neural Information Processing Systems*, pages 25224–25236, 2022. 2

[7] Julio Martin Duarte-Carvajalino and Guillermo Sapiro. Learning to sense sparse signals: Simultaneous sensing matrix and sparsifying dictionary optimization. *IEEE Trans. Image Process.*, 18(7):1395–1408, 2009. 2

[8] Michael Elad. Optimized projections for compressed sensing. *IEEE Trans. Signal Process.*, 55(12):5695–5702, 2007. 2, 4

[9] Shechao Feng, Charles Kane, Patrick A. Lee, and A. Douglas Stone. Correlations and fluctuations of coherent wave transmission through disordered media. *Phys. Rev. Lett.*, 61(7): 834–837, 1988. 3

[10] Isaac Freund, Michael Rosenbluh, and Shechao Feng. Memory effects in propagation of optical waves through disordered media. *Phys. Rev. Lett.*, 61(20):2328–2331, 1988. 3

[11] Nina M. Gottschling, Vegard Antun, Anders C. Hansen, and Ben Adcock. The troublesome kernel: On hallucinations, no free lunches, and the accuracy-stability tradeoff in inverse problems. *SIAM Review*, 67(1):73–104, 2025. 1

[12] Udith Haputhanthri, Andrew Seeber, and Dushan Wadduwage. Differentiable microscopy for content and task aware compressive fluorescence imaging. arXiv:2203.14945, 2022. 2

[13] Shaowei Jiang, Jun Liao, Zichao Bian, Pengming Song, Garrett Soler, Kazunori Hoshino, and Guoan Zheng. Axially shifted pattern illumination for macroscale turbidity suppression and virtual volumetric confocal imaging without axial scanning. *Opt. Lett.*, 44(4):811, 2019. 2

[14] Kyong Hwan Jin, Michael T. McCann, Emmanuel Froustey, and Michael Unser. Deep convolutional neural network for inverse problems in imaging. *IEEE Transactions on Image Processing*, 26(9):4509–4522, 2017. 2

[15] Ulugbek S. Kamilov, Charles A. Bouman, Gregery T. Buzzard, and Brendt Wohlberg. Plug-and-play methods for integrating physical and learned models in computational imaging: Theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 40(1):85–97, 2023. 4

[16] Michael R. Kellman, Emrah Bostan, Nicole A. Repina, and Laura Waller. Physics-based learned design: Optimized coded-illumination for quantitative phase imaging. *IEEE Transactions on Computational Imaging*, 5(3):344–353, 2019. 2

[17] Soumendu Majee, Thilo Balke, Craig A. J. Kemp, Gregery T. Buzzard, and Charles A. Bouman. Multi-slice fusion for

sparse-view and limited-angle 4D CT reconstruction. *IEEE Transactions on Computational Imaging*, 7:448–462, 2021. 2, 4

[18] A. Marian, F. Charrière, T. Colomb, F. Montfort, J. Kühn, P. Marquet, and C. Depeursinge. On the complex three-dimensional amplitude point spread function of lenses and microscope objectives: theoretical aspects, simulations and measurements by digital holography. *Journal of Microscopy*, 225(2):156–169, 2007. 3

[19] Eric Markley, Henry Pinkard, Leyla Kabuli, Nalini Singh, and Laura Waller. Computationally efficient information-driven optical design with interchanging optimization. arXiv:2507.07789, 2025. 2

[20] Elias Nehme, Daniel Freedman, Racheli Gordon, Boris Ferdman, Lucien E. Weiss, Onit Alalouf, Tal Naor, Reut Orange, Tomer Michaeli, and Yoav Shechtman. DeepSTORM3D: dense 3D localization microscopy and PSF design by deep learning. *Nature Methods*, 17:734–740, 2020. 2

[21] M. Pascucci, S. Ganesan, A. Tripathi, O. Katz, V. Emiliani, and M. Guillon. Compressive three-dimensional super-resolution microscopy with speckle-saturated fluorescence excitation. *Nature Communications*, 10:1327, 2019. 2

[22] Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11 (5–6):355–607, 2019. 5

[23] Henry Pinkard, Leyla Kabuli, Eric Markley, Tiffany Chien, Jiantao Jiao, and Laura Waller. Information-driven design of imaging systems. In *Advances in Neural Information Processing Systems*, 2025. 2

[24] B. Richards and E. Wolf. Electromagnetic diffraction in optical systems, II. structure of the image field in an aplanatic system. *Proceedings of the Royal Society of London. Series A*, 253(1274):358–379, 1959. 3

[25] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4):259–268, 1992. 2, 4

[26] Yoav Shechtman, Steffen J. Sahl, Adam S. Backer, and W. E. Moerner. Optimal point spread function design for 3D imaging. *Physical Review Letters*, 113(13):133902, 2014. 2, 4

[27] Kevin T. Takasaki and Jason W. Fleischer. Phase-space measurement for depth-resolved memory-effect imaging. *Opt. Express*, 22(25):31426–31433, 2014. 2

[28] Singanallur V. Venkatakrishnan, Charles A. Bouman, and Brendt Wohlberg. Plug-and-play priors for model based reconstruction. In *IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, pages 945–948, 2013. 2, 4

[29] Navodini Wijethilake, Mithunjha Anandakumar, Cheng Zheng, Peter So, Murat Yildirim, and Dushan Wadduwage. DEEP-squared: Deep learning powered de-scattering with excitation patterning. Zenodo, version 1. https://doi.org/10.5281/zenodo.8161051, 2023. 2

[30] Navodini Wijethilake, Mithunjha Anandakumar, Cheng Zheng, Peter T. C. So, Murat Yildirim, and Dushan N. Wadduwage. DEEP-squared: deep learning powered De-scattering with Excitation Patterning. *Light: Science & Applications*, 12:228, 2023. 2, 4

[31] Kyrollos Yanny, Nick Antipa, William Liberti, Sam Dehaeck, Kristina Monakhova, Fanglin Linda Liu, Konlin Shen, Ren Ng, and Laura Waller. Miniscope3D: optimized single-shot miniature 3D fluorescence microscopy. *Light: Science & Applications*, 9:171, 2020. 2

[32] Li-Hao Yeh, Shwetadwip Chowdhury, Nicole A. Repina, and Laura Waller. Speckle-structured illumination for 3D phase and fluorescence computational microscopy. *Biomedical Optics Express*, 10(7):3635–3653, 2019. 2, 3

[33] Hasan Yilmaz, Elbert G. van Putten, Jacopo Bertolotti, Ad Lagendijk, Willem L. Vos, and Allard P. Mosk. Speckle correlation resolution enhancement of wide-field fluorescence imaging. *Optica*, 2(5):424, 2015. 2

[34] Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7):3142–3155, 2017. 4